# Homework 2

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## 1. Question 1

*1.a.*

Yes, see reasoning in 1b.

*1.b.*

Define the payoff as $X$, price vector as $P$, and price of contingent claim as $p$. Solve the linear equation $Xp = P$ by elimination, we got

$$[X, P] \longrightarrow \begin{bmatrix} 1 & 0 & 0 & 0.2 \\ 0 & 1 & 0 & 0.4 \\ 0 & 0 & 1 & 0.2 \end{bmatrix} \tag{1}$$

Thus, we have complete markets in this economy and the price of continguent claim for state of nature 1, 2,3 are $[0.2, 0.4, 0.2]'$

*1.c.*

The price should be

$$[30, 20, 10] \times [0.2, 0.4, 0.2]' = 16 \tag{2}$$

*1.d.*

SDP in this economy is calucalteed by $M(i) = \frac{p(i)}{\pi(i)}$

$$M = [0.8, 0.8, 0.8]'. \tag{3}$$

This economy satisfies no-arbitrage as all the prices of contingent claims are strictly positive and unique. SDF remains the same accross all states, which means that agents in this economy are risk neutral.

*1.e.*

$$P_D = \sum_i \pi(i) M(i) X(i) = \frac{1}{4} \cdot 0.8 \cdot 30 + \frac{1}{2} \cdot 0.8 \cdot 20 + \frac{1}{4} \cdot 0.8 \cdot 10 = 16 \tag{4}$$

Homework
Solution

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## 2. Question 2

*2.a.*

| Parameter | Estimate | Std. error | t-stat |
|---|---|---|---|
| $a$ | 1.0368 | 0.0119 | 86.86 |
| $b_1$ (Lo 10) | −1.5850 | 1.6173 | −0.98 |
| $b_2$ (2-Dec) | 3.5503 | 1.8540 | 1.91 |
| $b_3$ (3-Dec) | 0.7104 | 2.0766 | 0.34 |
| $b_4$ (4-Dec) | −3.1122 | 1.8756 | −1.66 |
| $b_5$ (5-Dec) | 1.1837 | 2.0043 | 0.59 |
| $b_6$ (6-Dec) | 2.9972 | 1.9857 | 1.51 |
| $b_7$ (7-Dec) | −5.2931 | 1.8685 | −2.83 |
| $b_8$ (8-Dec) | 2.4722 | 1.6310 | 1.52 |
| $b_9$ (9-Dec) | 2.8775 | 1.6288 | 1.77 |
| $b_{10}$ (Hi 10) | −1.1771 | 1.0598 | −1.11 |

Two-step GMM, monthly, 1927:07–2026:08 (T = 1190). Excess returns in decimals. Newey-West S with automatic lag L = 19.

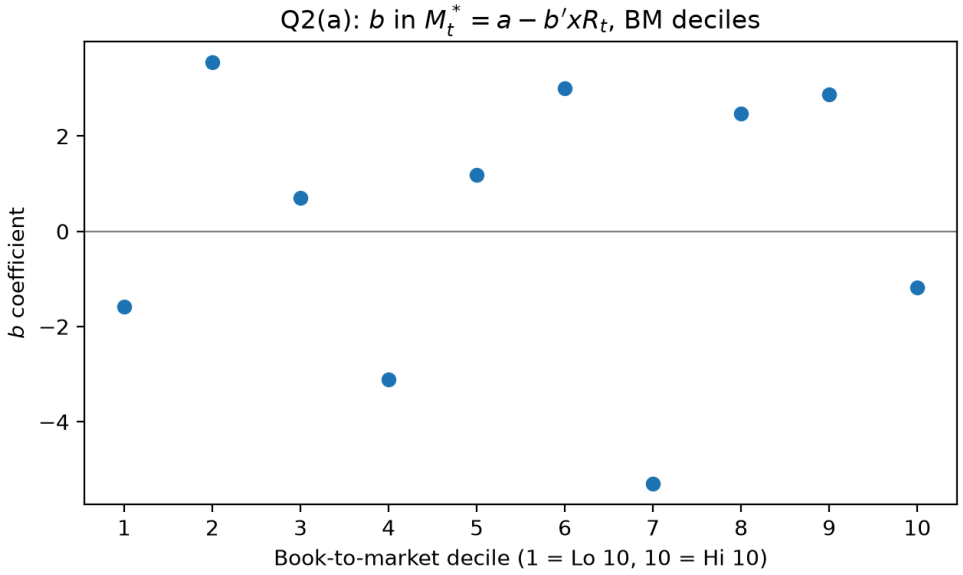



Figure 1: $b$ coefficients of $M_t^* = a - b'xR_t$ by book-to-market decile (two-step GMM, 1927:07–2026:08).

*2.b.*

| Parameter | Estimate | Std. error | t-stat |
|---|---|---|---|
| $a$ | 1.0666 | 0.0154 | 69.32 |
| $b_1$ (Lo PRIOR) | −4.0194 | 0.9747 | −4.12 |

| Parameter | Estimate | Std. error | t-stat |
|---|---|---|---|
| $b_2$ (PRIOR 2) | 1.1429 | 1.4347 | 0.80 |
| $b_3$ (PRIOR 3) | 0.1024 | 1.8005 | 0.06 |
| $b_4$ (PRIOR 4) | 1.4897 | 1.7155 | 0.87 |
| $b_5$ (PRIOR 5) | 2.5317 | 2.0073 | 1.26 |
| $b_6$ (PRIOR 6) | −0.8195 | 1.8827 | −0.44 |
| $b_7$ (PRIOR 7) | 0.2777 | 1.8301 | 0.15 |
| $b_8$ (PRIOR 8) | 0.9818 | 1.7812 | 0.55 |
| $b_9$ (PRIOR 9) | −0.9414 | 1.6701 | −0.56 |
| $b_{10}$ (Hi PRIOR) | 3.4370 | 0.8792 | 3.91 |

Two-step GMM, monthly, 1927:07–2026:08 (T = 1190). Excess returns in decimals. Newey-West S with automatic lag L = 3.

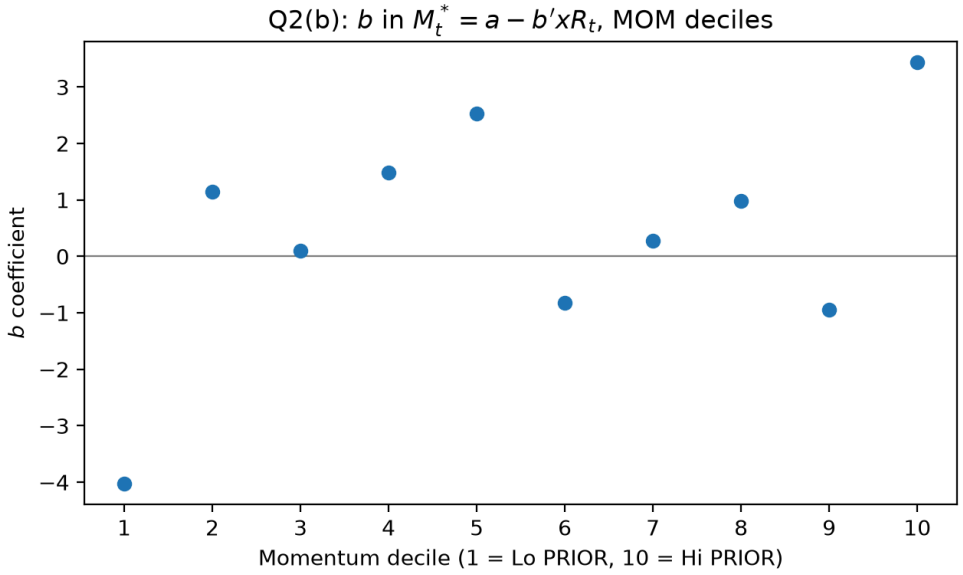



Figure 2: $b$ coefficients of $M_t^* = a - b'xR_t$ by momentum decile (two-step GMM, 1927:07–2026:08).

The sample correlation between $M_t^*$ from Question 2(a) (BM deciles) and $M_t^*$ from Question 2(b) (MOM deciles) is $\mathrm{Corr}\left(M_{BM,t}^*, M_{MOM,t}^*\right) = 0.3555$ (monthly, 1927:07–2026:08, $T = 1190$).

*2.c.*

| Parameter | Estimate | Std. error | t-stat |
|---|---|---|---|
| $a$ | 1.0927 | 0.0195 | 56.14 |
| $b_1$ (Lo 10) | −1.9664 | 1.8048 | −1.09 |
| $b_2$ (2-Dec) | 2.4739 | 1.8751 | 1.32 |
| $b_3$ (3-Dec) | 0.5781 | 2.1303 | 0.27 |
| $b_4$ (4-Dec) | −3.7439 | 1.8387 | −2.04 |

| Parameter | Estimate | Std. error | t-stat |
|---|---|---|---|
| $b_5$ (5-Dec) | 1.0655 | 2.1028 | 0.51 |
| $b_6$ (6-Dec) | 2.6840 | 1.7591 | 1.53 |
| $b_7$ (7-Dec) | −4.6456 | 1.7091 | −2.72 |
| $b_8$ (8-Dec) | 2.4707 | 1.6922 | 1.46 |
| $b_9$ (9-Dec) | 3.7398 | 1.6604 | 2.25 |
| $b_{10}$ (Hi 10) | −0.2415 | 1.0046 | −0.24 |
| $b_{11}$ (Lo PRIOR) | −4.4169 | 1.1081 | −3.99 |
| $b_{12}$ (PRIOR 2) | 0.6354 | 1.5542 | 0.41 |
| $b_{13}$ (PRIOR 3) | −0.1712 | 1.8918 | −0.09 |
| $b_{14}$ (PRIOR 4) | 1.9480 | 1.8068 | 1.08 |
| $b_{15}$ (PRIOR 5) | 1.4236 | 2.1083 | 0.68 |
| $b_{16}$ (PRIOR 6) | −0.8781 | 2.2007 | −0.40 |
| $b_{17}$ (PRIOR 7) | −0.1639 | 1.8900 | −0.09 |
| $b_{18}$ (PRIOR 8) | 0.9802 | 1.8749 | 0.52 |
| $b_{19}$ (PRIOR 9) | −1.2708 | 1.8496 | −0.69 |
| $b_{20}$ (Hi PRIOR) | 3.4473 | 1.0952 | 3.15 |

Two-step GMM, monthly, 1927:07–2026:08 (T = 1190). Excess returns in decimals. Newey-West S with automatic lag L = 7.

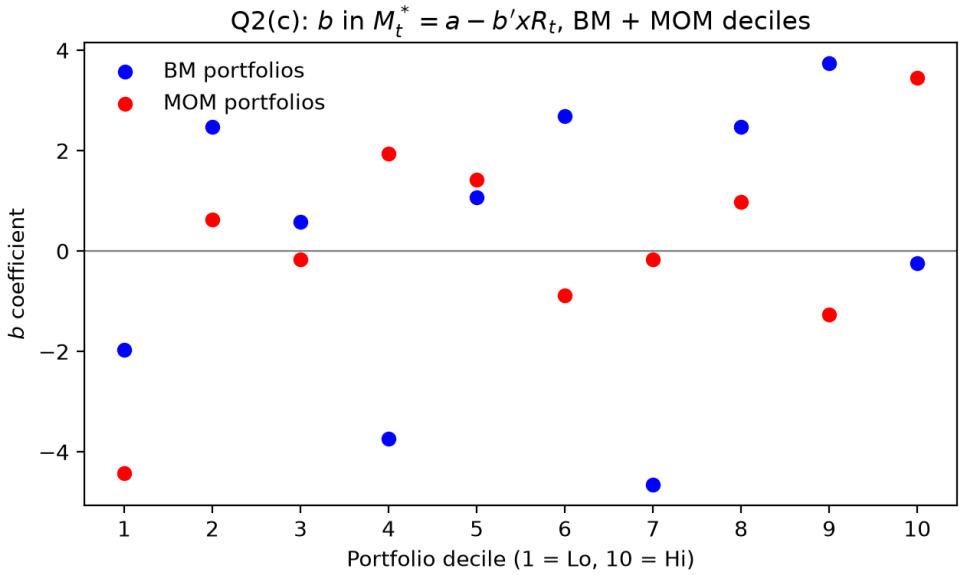



Figure 3: $b$ coefficients of $M_t^* = a - b'xR_t$ estimated on the 20 BM and MOM decile portfolios jointly, by decile (blue: BM portfolios; red: MOM portfolios; two-step GMM, 1927:07–2026:08).

| Question | Test assets | Annualized Sharpe ratio of $xR_t^* = b'xR_t$ |
|---|---|---|

| Question | Test assets | Annualized Sharpe ratio of $xR_t^* = b'xR_t$ |
|---|---|---|
| 2(a) | 10 BM deciles | 0.6646 |
| 2(b) | 10 MOM deciles | 0.8937 |
| 2(c) | 20 BM + MOM deciles | 1.0545 |

Monthly 1927:07–2026:08 (T = 1190); Sharpe ratio = $\sqrt{12}$ × mean / s.d. of the monthly $xR_t^*$ series.

## 3. Question 3

*3.a.*